DEPLOYMENT REPORT

Student Service Request Monitoring System

Project Overview

The Student Service Request Monitoring System is a Streamlit decision support system that helps users to explore service request data, display data visualizations, and generate a Decision Tree prediction regarding an SLA violation. The user interface groups the activities according to Home, Dataset, Data Analysis, Prediction, Dashboard, and System Design pages.

Reused Code and Data

The project is built on the basis of the Google Colab notebook created for the midterm and the provided synthetic service request data set. The preparation and prediction methods have been converted into modules to allow the Streamlit app to load, prepare, and predict using the saved model. In addition, the project takes advantage of the Decision Tree method used in the notebook.

Changes Made

- Designed an application using Streamlit where pages have been designed for data visualization, data analysis, data prediction, dashboard summary, and system design.
- Split processing code, exploratory analysis code, prediction code, and dashboard code into separate Python modules.
- The application includes a saved model artifact along with model metadata.
- A requirements file has been included for package installation along with a README file explaining local execution of the application.
- A use case diagram and decision support workflow have been provided to explain the application's usage.

Testing and Deployment

The application was run locally using Streamlit and accessed from a web browser. When deploying in the cloud, the import error occurred due to the absence of the Python files in the modules folder. Repository setup was corrected by moving all the files in modules. Once this is done, the app needs to be re-deployed and each page tested. The app is considered to be successfully deployed only when the live URL opens and the pages work.

Steps to follow for the final testing of the app: open the live app and access Home, Dataset, Data Analysis, Prediction, Dashboard, and System Design pages; provide at least five different inputs for predictions and check the output; and ensure that the use case diagram and workflow can be seen.

Limitations

This data is synthetic and is to be used for academic purposes only. The model's performance on the synthetic test data does not necessarily imply its performance on actual student service requests. Predictions made by the model may not be accurate, and thus, can only serve as a tool for decision-making. The original student service request should be consulted before making any decisions.

Reflection

This project proved that it is possible to turn a workflow for mining data into a user interactive decision support application. The use of modular coding made the project easier to manage, and the use of Streamlit made it easy to present tables, graphs, and predictions. It is important for deployment purposes to ensure that the Python import statements match the directory structure of the project repository. In future works, this system should be improved through testing with real data, evaluations, and user feedback.

Links for Submission

GitHub repository: https://github.com/kitchieminez9-cpu/Student-Service-Monitoring

Streamlit deployed URL: https://student-service-monitoring.streamlit.app/

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